

The Chicago Blackhawks

The Chicago Blackhawks, one of the NHL’s “Original Six” teams founded in 1926, operate a large retail business that sells team merchandise through the United Center, online channels, and other locations. Because merchandise demand rises and falls throughout the season, planning the right amount of inventory is both financially and operationally important. Ordering too much inventory ties up capital and storage space, while ordering too little can lead to missed sales when popular items sell out during high-demand games. To address this challenge, the Chicago Blackhawks partnered with the University of Chicago’s Data Science Clinic to improve how merchandise sales are forecasted.

The team developed an interactive dashboard that estimates merchandise sales across different types of purchases and time periods, allowing users to explore expected demand and plan inventory accordingly (Figure 1). The tool also measures uncertainty in the forecasts and provides a confidence interval that reflects the range of possible outcomes. Users can adjust the confidence level depending on how conservative they want their inventory planning to be. By combining forecasts based on historical sales patterns with game schedule and retail activity data, the dashboard helps decision-makers better anticipate demand and manage merchandise inventory throughout the season.

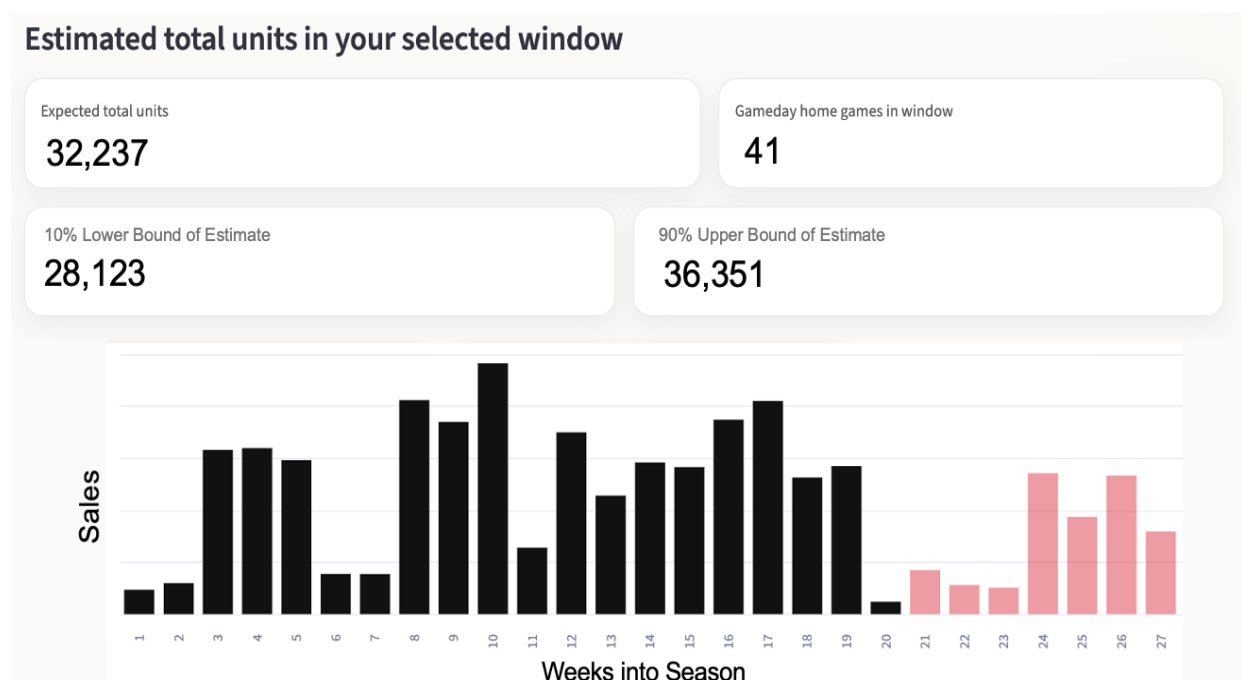


Figure 1. Sales forecasting dashboard with confidence interval. Black bars show observed merchandise sales during the season, while red bars represent forecasted sales for the remaining weeks.

*Numbers and figures are merely illustrative and do not represent actual sales numbers and rates.